

RESEARCH PROPOSAL

STANDARDISATION OF DIGITAL FORENSIC PROTOCOLS AND ESTABLISHING RELIABILITY METRICS FOR LEGAL REPORTS

Group Members

1. Wan Wei Siang – 251UC2508W
2. Elsa Zara Binti Fakhurrazi – 251UC2502P
3. Terry Teoh Jay Poh – 251UC2518F
4. Nur Afqah Ramli – 251UC2514E

Programme: Research Method

Course: CPT6123

Lecturer: Rohana Binti Mydin

Semester: 2620

Submission Date: 29 September 2026

Abstract

Digital evidence is now central to prosecuting cybercrime, yet the field of digital forensics lacks standardised procedures and quantifiable measures of evidential reliability. An empirical analysis of 40 digital forensic reports found that most described their methods vaguely, none provided statistical support for reliability or validity, and most acknowledged limitations without quantifying error rates. This makes findings hard to reproduce and difficult to defend under legal admissibility standards such as the Daubert criteria. This proposal addresses that gap through four objectives: (1) a systematic review of current forensic methodologies and their documented deficiencies; (2) the design of a unified protocol covering identification, collection, preservation, examination, analysis and reporting, refined through expert review using the Delphi method; (3) the development of a reliability metric that scores digital evidence using factors such as the acquisition method, volatility, temporal decay, chain-of-custody integrity and tool validation status; (4) a controlled empirical evaluation comparing the standardised protocol against ad hoc baselines across simulated forensic scenarios. Performance will be measured using reliability scores, Cohen's kappa, error rate, F1-score, chain-of-custody integrity rate and latency, with statistical significance testing. This study anticipates a validated protocol and metrics that improve reproducibility and give courts a transparent, objective basis for weighing digital evidence.

Contents

1	Introduction	3
2	Literature Review (Recap)	4
3	Problem Statement	6
3.1	Context	6
3.2	Gap	6
3.3	Solution	7
4	Research Questions	8
5	Review Objectives	9
5.1	Objective 1 (Addresses RQ1)	9
5.2	Objective 2 (Addresses RQ2)	9
5.3	Objective 3 (Addresses RQ3)	10
5.4	Objective 4 (Addresses RQ4)	10
6	Research Methodology	12
6.1	Research Design	12
6.2	Technical Approach	12
6.2.1	Evidence Type	12
6.2.2	General Principles	14
6.2.3	Protocol Layers	15
6.3	Data Tools	20
6.3.1	Selection justification	21
6.4	Evaluation Criteria	21
6.4.1	Assumptions & Dependencies	22
6.4.2	The Delphi Method	22
7	Expected Outcomes	23
8	Project Timeline	24
8.1	Conclusion	25
	References	26

Chapter 1

Introduction

The present state of computer science allows for constant advancement in technology within the modern world. Of many innovations, AI has become a central subject within cybersecurity, for its rapid improvement of software through constant iteration and self-progression.

However, this ability to rapidly improve is an emerging problem, as the sophisticated attacks; deepfakes, phishing, automation of malware, combined with this age of interconnected devices, the Internet of Things (IoT) give rise to more complications in updating the defenses against these attacks.

As such, traditional methods for combating modern cybercrimes prove to be ineffective and outdated. According to Sharma, 2024, there is a large empirical gap within the research in digital forensics. Its empirical analysis of 40 digital forensic reports presents results with 82.5% vague descriptions in their research methods, with 0% providing proper literature statistics for credibility and 90% acknowledging limitations in their paper.

Furthermore, the lack of standardization in digital forensics creates difficulties in cases pertaining to cybercrime, as evidence can be insufficient, its procedure of collection unrecognized in legal systems, as well as make it difficult for validity in international cases. This highlights the importance of acknowledging that research in digital forensics is in its infancy, and in need of proper analysis.

In light of these issues, this report aims to propose a standard protocol in gathering digital evidence and the establishment of reliability metrics to assist legal cases by removing uncertainty, introducing objectivity, and increasing the reliability of evidence.

Chapter 2

Literature Review (Recap)

From the literature that we've analysed during Assignment 1, there appears to be several broad consensus in regards to digital forensics and modern cybercrime:

- **The essentiality of digital forensics and the insufficiency of the traditional methods in combating cybercrime.** Research consistently indicates that digital forensics is vital for investigating and prosecuting cybercrimes through assisting law enforcement officers in locating, examining and protecting electronic data that can be used as digital evidence in court to identify suspects and establish legal facts. (Arif et al., 2025, Ilyas et al., 2024)

On the other hand, traditional forensic techniques are not adequate to combat contemporary cybercrime. The sophistication of cybercrime methodology utilised by perpetrators such as encryption and anti-forensic softwares that are able to conceal their activities from relevant investigators while allowing malicious actors to evade the standard security filters; large data volumes which, when analysed through traditional means, is slow and at times impossible due to the size of the data volume are clear signals that the traditional methods are insufficient and at times detrimental against combating cybercrime, as events like security breaches can cause massive damage before investigators are able to identify them through the traditional methods.

- **Lack of standardisation in digital forensic methodologies.** Research articles obtained highlight a clear lack of standardised protocols across the field of digital forensics. Different investigators frequently apply divergent methodologies, which often leads to inconsistent analytical outcomes in resultant research articles and reports. (Ilyas et al., 2024, Arif et al., 2025, Sharma, 2024)
- **Possible upgrades towards digital forensics through AI and Zero Trust Architecture.** As cybercriminals have utilised AI software for generating steganographic code to mask hidden information while they commit cybercrimes (Osamor et al., 2026, Sanna et al., 2025), various articles have suggested the implementation of artificial intelligence (AI), machine learning and deep learning within digital forensics as aid against said AI-based attacks. Through proper utilisation of AI as an assistant in digital forensics, systems are able to analyse large datasets in a short time, automate dull and repetitive duties and identify possible configurations that human investigators may overlook. This allows professionals to operate with much greater speed and precision in comparison.

In terms of Zero Trust, a couple of research articles proposed utilising a Zero Trust Architecture, which operates on a “never trust, always verify” mantra (Prajwalasimha et al., 2025, Arif et al., 2025). This allows for a much more effective protection mechanism against complex cyber threats and deceptive email campaigns. Besides that, they also suggest implementing a layered defense strategy as well, to better filter out cyberthreats alongside providing safe and secure protection against critical assets.

Chapter 3

Problem Statement

3.1 Context

Present-day Investigators specializing in digital forensics have a critical role in providing assistance to law enforcement in regards to digital crimes, particularly ones involving computers. These specialists influence how electronic data is collected, examined and stored for legal cases, determining how attorneys may approach or strategize their court proceedings.

As crimes on the internet become increasingly complex, such as advanced persistent threats (APTs), ransomware, deepfakes and methods that disrupt or even prevent investigations, legal systems will consequently depend heavily on digital data as evidence.

In many nations, judges follow a standard such as the Daubert criteria (Carranza and De-Cusatis, 2021). As such, it is expected that the courts require specialists to show their method of gathering data has specified measured mistakes, have undergone checks by other scientists, and ultimately approved by the scientific community.

3.2 Gap

The field of digital forensics lacks uniform procedures for its operations. According to Sharma, 2024, where a study of 40 digital forensic reports was made, researchers found multiple specific weaknesses. The findings reveal that 33 of those documents used unclear language to describe how investigators performed their tasks. There is no instance where the authors included mathematical data to prove consistency or accuracy in their results.

Despite 36 reports addressing flaws through noting constraints existing in their studies, the specific quantitative data measuring the frequency of mistakes were not included in their articles.

As previously indicated, the lack of fixed rules causes different specialists to reach different conclusions when they examine identical datasets. As a result, the process of repeating investigations to verify results proves difficult for investigators to achieve.

Considering these factors, legal systems may refuse to accept digital evidence as viable proof. Paradoxically however, there does not exist a system to measure the trust a user can place in digital data. This information is crucial in providing the framework for specialists to gather data, such as the storage logs remaining intact and the details on when events occurred.

Additionally, the legal system is currently without concrete rules to determine how much importance a judge should give to this evidence.

3.3 Solution

We propose a process where investigators use a uniform digital forensic protocol and a system that measures reliability through quantitative data to evaluate digital evidence. The protocol describes the required steps during every forensic stage, including how investigators identify, collect, preserve, examine, analyze and report data. Moreover, we incorporated a reliability metric that assigns a numerical score between one and ten based on factors that researchers have validated through observation.

The factors as mentioned include the method used to acquire data, the rate at which data changes, the time taken before recovery as well as the degree of completion in the chain of custody.

By creating a framework that experts can successfully repeat and explain clearly, it is anticipated that uncertainty will be removed, resulting in legal systems being able to admit digital evidence based on facts. Ultimately, this paper aims to create a method for electronic files to be consistent and neutral to ease the process of legal evaluations on computer crimes.

Chapter 4

Research Questions

The 4 Research Questions for this specific Research Proposal is as follows:

RQ1: There are many current commonly used digital forensic techniques. What are they and what are the flaws in the documentation and consistency of these techniques?

RQ2: What is required in a standard digital forensic protocol that would enhance consistency, repeatability and court admissibility of digital evidence in a variety of forensic cases?

RQ3: What makes digital evidence reliable and how can this be quantified for use within the court system to evaluate the reliability of digital evidence?

RQ4: How does adopting a standard forensic protocol and a reliability system enhance the repeatability of forensic analysis and the admissibility of digital evidence within a mock court setting?

Chapter 5

Review Objectives

5.1 Objective 1 (Addresses RQ1)

To systematically identify, analyse and document the current commonly used digital forensic techniques and the flaws in their documentation and consistency by conducting a comprehensive systematic literature review of peer-reviewed studies published between 2021 and 2026, extracting procedural steps, documentation practices, consistency and repeatability issues, and reported limitations across at least 50 relevant studies within the first six months of the project.

SMART Criteria

- **Specific:** Identify commonly used digital forensic techniques and analyse flaws in documentation and consistency.
- **Measurable:** Review minimum 50 peer-reviewed articles; extract and categorise techniques, procedural steps, documentation practices, consistency indicators, and stated limitations.
- **Achievable:** Access IEEE Xplore, ScienceDirect, and ACM Digital Library through institutional subscriptions.
- **Relevant:** Establishes the empirical foundation for identifying what a standardised protocol must address.
- **Time-bound:** Completed within Months 1–2 of the project timeline.

5.2 Objective 2 (Addresses RQ2)

To design and develop a standardised digital forensic protocol that integrates best practices from existing frameworks (NIST, ACPO, ISO/IEC 27037) into a unified, phase-by-phase procedure with mandatory documentation requirements, explicitly structured to enhance consistency, repeatability and court admissibility of digital evidence across a variety of forensic cases, validated through expert review by at least five certified digital forensic practitioners and achieving $\geq 80\%$ expert consensus via the Delphi method within nine months of project commencement.

SMART Criteria

- **Specific:** Create a comprehensive protocol covering identification, collection, preservation, examination, analysis, and reporting, including documentation templates, repeatability controls, and admissibility mapping.
- **Measurable:** Protocol must include minimum 40 discrete procedural steps; achieve $\geq 80\%$ expert consensus via Delphi method; validated by at least five certified practitioners.
- **Achievable:** Based on established standards and iterative expert feedback.
- **Relevant:** Directly addresses the standardisation, consistency and admissibility gaps identified in RQ1 and RQ2.
- **Time-bound:** Draft completed by Month 4; expert validation completed by Month 10.

5.3 Objective 3 (Addresses RQ3)

To formulate and validate a reliability metric system for digital evidence that defines the factors of making digital evidence reliable and assigns quantifiable scores (1–10 scale) based on empirically derived weightings for the acquisition method, evidence volatility, temporal delay, chain of custody integrity, and tool validation status, tested against a corpus of at least 100 historical forensic cases to establish scoring accuracy and inter-rater reliability (Cohen’s $\kappa \geq 0.80$) and to assess its usability within the court system within twelve months of project commencement.

SMART Criteria

- **Specific:** Develop a numerical scoring system for digital evidence reliability and define the underlying reliability constructs for court evaluation.
- **Measurable:** Validate against 100 cases; achieve Cohen’s $\kappa \geq 0.80$ for inter-rater reliability; document scoring accuracy compared to expert consensus; assess court usability.
- **Achievable:** Historical case data available through law enforcement partnerships and public court records.
- **Relevant:** Provides courts with objective criteria for assessing evidentiary weight and reliability.
- **Time-bound:** Metric developed by Month 10; validation completed by Month 11.

5.4 Objective 4 (Addresses RQ4)

To empirically evaluate the effectiveness of the proposed standardised protocol and reliability metric through a controlled experimental study comparing forensic outcomes (reliability scores, reproducibility via Cohen’s Kappa, error rates, and processing latency) between the standardised protocol and baseline ad hoc methodologies across at least 20 simulated forensic scenarios, and through a mock court setting assessing the repeatability of forensic analysis and the admissibility of digital evidence, with results demonstrating statistically significant improvement ($p < 0.05$) in reproducibility and error reduction within eighteen months of project commencement.

SMART Criteria

- **Specific:** Conduct controlled experiments comparing standardised vs. ad hoc protocols, plus a mock court assessment of repeatability and admissibility.

- **Measurable:** 20+ scenarios; statistical significance at $p \leq 0.05$; improvement in Cohen's κ and error rate reduction documented; mock court admissibility ratings collected.
- **Achievable:** Utilises established forensic datasets, controlled laboratory conditions, and mock court participants such as legal practitioners or mock judges.
- **Relevant:** Provides empirical evidence for protocol efficacy, repeatability, and legal defensibility.
- **Time-bound:** Experimental design finalised by Month 10; data collection completed by Month 5; mock court assessment and analysis completed by Month 11.

Chapter 6

Research Methodology

6.1 Research Design

In terms of Research Design, we've elected to use an empirical evaluation design through the methodology of controlled and repeated measures. The methodology will require forensic datasets, which will be used to be processed through the proposed standardised protocol alongside the baseline ad hoc protocols, while using constants to ensure that the protocol is the only systematically varied factor besides from the type of evidence obtained from the dataset. Afterwards, each condition obtained from the processed protocols will be run through independent trials from varying pipeline instances to support significance testing alongside providing multiple evidence samples per category to avoid cramming all the conclusions into a single case.

The dependent variables for this methodology would be the reliability score for each protocol, the reproducibility of the results amongst analysts (through Cohen's κ), integrity of the evidence from the chain of custody, time taken for protocols to produce a report and the error rate in comparison to the ground truth.

The main reasoning behind using empirical evaluation is that in accordance to the reliability queries under the Daubert standard, a legal report reliability claim is required to have a known, repeatable and quantifiable error rate alongside the procedure being tested, peer-reviewed, published and generally accepted by the relevant community, in this situation being the Digital Forensics field (Carranza and DeCusatis, 2021). Through the use of a controlled empirical design and by the nature of the protocol being the sole variable in said environment, we are able to obtain a known error rate with statistical confidence. In terms of using other designs, the ground truth cannot be obtained from methods such as analysing the data obtained from live caseworks as there is no independently verifiable correct answer from said cases, hence an error rate is unable to be measured using said design and it can be deemed that empirical evaluation is the best suited for this specific research.

6.2 Technical Approach

6.2.1 Evidence Type

The following table outlines the scope of the digital forensic domains to be tested against the proposed standardized protocol, with a focus on corresponding evidence types.

Table 6.1: Evidence Types

Domain	Evidence Type	Example Findings
Disk	Forensic disk image	1.Deleted Files 2.File metadata 3.Documents
Network	PCAP/network capture	1.Port scans 2.Phishing traffic 3.Suspicious DNS requests
AI-generated Media	Image/video/audio	1.Deepfake 2.Synthetic media

The following table shows the general procedure of executing the research to compare the proposed standard protocol with the ad hoc method.

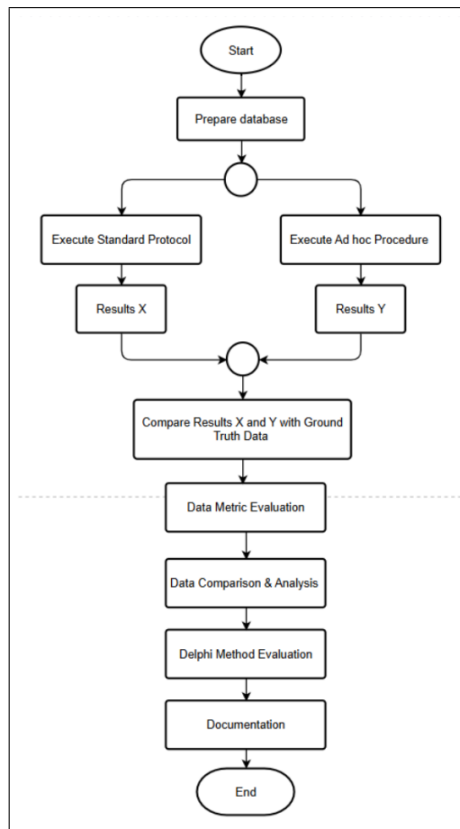


Figure 6.1: Experiment General Procedure

The following table summarises the proposed protocol layers for the standardization of forensic procedures.

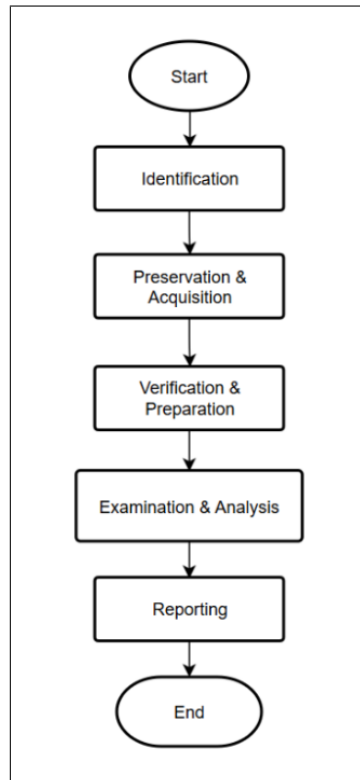


Figure 6.2: Protocol Layers

6.2.2 General Principles

This section outlines the general principles that must be followed when the investigation is carried out:

1. The evidence must immediately be seized upon discovery.
2. The original evidence must not be tampered or directly examined where it may alter its initial state.
3. The evidence to be examined by forensic tools must only be examined by its copy, and never the original source. This prevents data overwriting in volatile memory.
4. If access to original data cannot be avoided, every action must be recorded in its investigation.
5. The investigation must be replicable by a third party.
6. All actions made in the investigation must be recorded through a reliable logging system.
7. Throughout the investigation, an audit trail must be maintained on all actions done on the evidence, whether original or copied. The details related to the trail must be recorded extensively, such as the investigator carrying out the action, the date, time, etc.

6.2.3 Protocol Layers

The following section further expands each protocol and provides a checklist to be filled in every investigation.

Protocol 1: Identification

By establishing the facts and the general details of the case, the investigators will form an accurate context of the case, preventing confusion or misinterpretation of data. Additionally, by knowing the state of the evidence, investigators will be able to formulate a plan to further examine the evidence in the best strategy to minimize data loss. Before the investigation is carried out, the investigator must record the following into a document:

- ☐ Record investigator name
- ☐ Record case ID
- ☐ Record device owner details
- ☐ Record general case facts
- ☐ Record date and time received
- ☐ Record evidence sender details
- ☐ Record evidence source location
- ☐ Record date and time of evidence seizure
- ☐ Record method of evidence collection

The investigator must record the following facts of the evidence:

- ☐ Assign evidence ID
- ☐ Record evidence type
- ☐ Record device type
- ☐ Record any visible applications/notifications if applicable
- ☐ Record device initial state (locked/unlocked/etc)
- ☐ Record device network connection
- ☐ Record device physical condition and/or damages
- ☐ Photograph device in proper lighting/angles
- ☐ Record preliminary context of evidence to the case

Protocol 2: Preservation & Acquisition

The seized evidence must follow a strict procedure to prevent data loss, which may lead to evidence insufficiency and loss of crucial evidence.

The investigator must adhere to the following procedure for evidence preservation:

- ☐ Establish and record the chain-of-custody
- ☐ Record evidence storage conditions
- ☐ Develop a plan to protect evidence from physical damage
- ☐ Prevent unauthorised access
- ☐ Maintain state of device (ie: battery life, power status...)
- ☐ Extensively document any interactions with original evidence

The data from the evidence may be acquired by many different sources. Data acquiring must be carefully planned to prevent as much data loss where possible, executed with the least amount of effort required.

To develop a plan, the investigators must consider these factors that may influence how the data is acquired:

1. **Volatility:** Volatile data is data that may be lost after the device has been switched off, and also may be lost from other tasks performed on the device. This data must be given the priority over non-volatile data.
2. **Potential data loss:** Investigators must record and determine the potential data losses that may occur when data is acquired.
3. **Forensic Tool:** With the information gathered on the device, investigators must determine the most suitable forensic tool to extract data from the evidence.

The following is the procedure required to develop a data acquisition plan.

- ☐ Identify acquisition objectives
- ☐ Determine method of acquisition
- ☐ Determine whether acquisition on original evidence must be required
- ☐ Identify data sources
- ☐ Assess data volatility
- ☐ Assess potential data loss
- ☐ Develop risk assessment
- ☐ Select appropriate forensic tool for acquisition
- ☐ Record tool names, versions and any configurations
- ☐ Develop alternative plans to carry out investigation

Once a plan is created for data acquisition, the following steps must be executed.

- ☐ Create forensic image/copy of evidence with chosen forensic tool if applicable
- ☐ Acquire forensic data with the chosen acquisition method if applicable
- ☐ Record and log all copies of the evidence

Protocol 3: Verification & Preparation

For the development of a case, it is crucial to be able to explain every step that occurs when data is acquired, this is to prove that investigators did not tamper with the evidence, ensuring data integrity for admissibility in legal systems.

Once data from the evidence is acquired, the investigators must execute the following to verify and prepare the data for examination and analysis:

- ☐ Run SHA-256 hash function on original data and copied data if applicable, alternatively, generate any form of cryptographic hash value of each.
- ☐ Record hash algorithm
- ☐ Record date, time and any relevant details of the hashing
- ☐ Record hash value in evidence log
- ☐ Compare hash
- ☐ Record verification result and timestamp
- ☐ Secure all data from evidence
- ☐ Prepare chosen verified data for examination
- ☐ Secure perimeter of device

In the case that the verification on the data integrity is compromised, the discrepancy is documented, its cause is investigated, and the method of acquisition is repeated using the appropriate or alternative method where applicable.

Protocol 4: Examination & Analysis

The examination stage of the investigation must be done carefully to ensure the data is thoroughly assessed for analysis.

The examination of data requires a development of a suitable plan. The following lists out the factors to consider when executing data examination.

1. **Identification of relevant data:** The investigator must identify which data sources, applications, system records, metadata, files and others that are relevant to the investigation to be examined.
2. **Data filtration:** Forensic tools must be identified to filter data and extract only relevant data through a predefined criteria. The criteria to be established may include file type, user details, timestamps, search keywords, application, or other characteristics which ensures reproducibility.

3. **System bypassing & data-access barriers:** Data/code may be hidden behind features provided by certain software, such as the system's OS and applications or the data may be behind a password or encrypted. Thus, investigators must create a strategy to overcome such systems using known techniques.
4. **Forensic tool & techniques selection:** The forensic tool selected by investigators must be appropriate to the context of the evidence, and the examination objectives.

The following is the procedure required to develop a data examination plan.

- ☐ Define examination questions
- ☐ Identify relevant data types and sources
- ☐ Establish data filter criteria
- ☐ Identify examination limitations
- ☐ Identify potential data-access barriers
- ☐ Strategize overcoming data-access barriers
- ☐ Select suitable forensic tools/techniques
- ☐ Record all forensic tool details (names, versions and configurations)
- ☐ Record procedures for executing Examination
- ☐ Record all details pertaining to the examination

Once the examination plan has been defined, the investigator must execute the plan through a systematic, controlled approach, with appropriate documentation. There are several factors to consider when carrying out the analysis.

1. **Data access & recovery:** When accessing data using the selected forensic tool, investigators must identify the data that are inaccessible, with proper documentation of its recovery and the techniques used.
2. **Data filtering:** Data is filtered through an established criteria made by the investigation to ensure resources are not spent on irrelevant data. This process must be done carefully, as to not exclude evidence that may be crucial. All data that are excluded must be documented accordingly.
3. **Individual data analysis:** Individual data properties are examined to determine its content, timestamps, metadata, any other relevant information to the investigation.
4. **Data relationship analysis:** Investigators must identify relationships between information, which may support conclusions made.
5. **Comparison and contradictions:** Findings made from different techniques are compared and searched for any contradictions, which may warrant re-evaluation.
6. **Critical Analysis:** Investigators must critically evaluate the obtained findings and relate evidence with one another. Resulting in creating conclusions with supporting evidence. Additionally, investigators must ensure that their reasoning behind conclusions considers alternate possibilities. As such, all interpretations must be recorded, and having a clear distinction between fact from observation, with interpretation.

7. **Documentation of uncertainty:** Any limitations by the forensic tools, or any data inaccessibility, or uncertainties in any findings have to be documented extensively. This is to ensure any information is not excluded when building a defensible forensic case.

The following procedures are required to be executed for the examination and analysis.

- ☐ Access data through selected method
- ☐ Record data access or data recovery
- ☐ Extract relevant information
- ☐ Filter irrelevant data
- ☐ Record excluded data and document reasonings
- ☐ Analyse individual data and corresponding metadata
- ☐ Examine data relationships
- ☐ Identify contradictions
- ☐ Compare information
- ☐ Record observations
- ☐ Record interpretation of findings
- ☐ Evaluate findings critically
- ☐ Record viable correlations
- ☐ Document uncertainties

The findings made by the examinations and analyses of data must be verified through repeating the procedures, or by using alternate forensic tools. Any contradictions must be resolved before documentation.

Protocol 5: Reporting

The synthesizing of a concise, comprehensive report of the investigation's findings is crucial for presenting evidence of the case, as well as giving a detailed log of every action in the case to prevent misinterpretation.

The following outlines the sections required to be included into the formal documentation.

1. **Case information:** All records, such as the IDs, details and names of associated people, device and evidence regarding the case must be included.
2. **Scope:** The investigators must outline the specific scope and objections of the investigation.
3. **Forensic methodology:** The details that must be included are as follows:
 - Details of forensic tools used (ie: tool name, tool version)
 - Settings/configurations of device

- State of device
 - Part of evidence examined
 - Any related procedures in the data examination
4. **Evidence integrity:** the documentation must include the hash values calculated, the record of copies made in the investigation, the information regarding the chain-of-custody, and the results of the evidence integrity verification.
 5. **Synthesized findings:** The results observed, supporting artefacts, interpretations and analyses must be recorded.
 6. **Limitations:** Document the unrecoverable data of the case, missing evidence, tool limitations, uncertainty in interpretation, and any details that may limit an aspect of the case.

When corroborating information and creating the report, it is imperative to ensure that there is a clear separation between the observations/facts of the case and the interpretations made by the investigator.

The following lists the procedure following the documentation process.

- ☐ Verify IDs/names are accurate
- ☐ Verify all forensic tools used are included
- ☐ Verify that the findings are supported by real evidence
- ☐ Verify the comprehensiveness of the report
- ☐ Verify all major procedures are recorded
- ☐ Ensure any references are included
- ☐ Verify format consistency of the report
- ☐ Conduct a document review from a third party
- ☐ Record final details of the document (ie: report version, finalization date)

6.3 Data Tools

The following table details the primary forensic tools and datasets to be used when applicable to the forensic investigations in our proposed research.

Table 6.2: Data Sources And Data Tools

Domain	Evidence Type	Example Findings	Dataset sources	Primary forensic tools examples
Disk	Forensic disk image	1.Deleted Files 2.File metadata 3.Documents	1. NIST CFReDS forensic disk image 2.Digital Corpora Repository	1.Autopsy 2.FTK Imager 3.Guymager
Network	PCAP/network capture	1.Port scans 2.Phishing traffic 3.Suspicious DNS requests	1.NETRESEC Public PCAP repository 2.Malware Traffic Analysis	1.Wireshark 2.Arkime 3.Suricata 4.Zeek
AI-generated Media	Image/video/audio	1.Deepfake 2.Synthetic media	1.FaceForensics++ 2.TrueFake 3.Kaggle	1.ExifTool 2.Reality Defender 3.Resemble AI (DETECT-2B) 4.Sensity AI

6.3.1 Selection justification

The selected dataset represents the array of various types of digital evidence that may be included in forensic investigations that can be analysed against the proposed standard protocol. The scope of our evidence is the disk, network, and synthetic media. The selection was made as it provided a large dataset for thorough testing of the protocol and for evaluation of the forensic tools. Furthermore, compatibility and reproducibility were considered, through selection of data in the correct evidence format. Each dataset provided ample varying characteristics that allows for proper evaluation of the protocol.

6.4 Evaluation Criteria

To ensure the resulted outcome from the protocols are able to hold up against legal scrutiny, we deem it necessary to utilise the metrics shown in Table X in order to better showcase the protocols functionalities in three distinct factors: Is the protocol developed compliant with its reliability, can the resultant outcome be reproduced and how precise and accurate is said outcome.

Table 6.3: Metrics used for Evaluating the Protocols

Metric	Definition	Factor Showcased
Reliability Score	Primary outcome in comparison to protocol condition (Range of 0 to 1)	Is the protocol compliant with its reliability
Cohen’s Kappa coefficient (κ)	Reproducibility of outcome using identical evidence/dataset ($\kappa \geq 0.8$, strong agreement in reproducibility)	Can the resultant outcome be reproduced
Chain of custody Integrity Rate	Rate of verification between hash chain nodes without any breaks between	Is the protocol compliant with its reliability
Latency (ms or hours)	Wall clock time taken for each pipeline stage (end to end)	Is the protocol compliant with its reliability
Error Rate	(False Positive + False Negative findings) / Ground Truth findings	How precise is the outcome
F1 score	Harmonic mean of precision and recall $[(2 * \text{Precision} * \text{Recall}) / (\text{Precision} + \text{Recall})]$	How accurate is the outcome

6.4.1 Assumptions & Dependencies

As previously mentioned, our research is focused on executing the standard protocol against a known dataset to be compared with the ground truth. However, this is under the assumption that the steps from protocol 1,2, and 3 have been executed, providing data suitable for the procedures of examination & analysis and the protocols thereafter.

6.4.2 The Delphi Method

After obtaining the evaluated data from the standard protocol and the ad hoc procedure; the results are further analysed by the established approach, the Delphi method, known for its accuracy in judging a solution to a problem and its successful results, according to the (insert source). The aim is to gather unbiased expert opinion on the results to determine which of the two procedures are preferred. Generally, the method is a qualitative approach with many iterations to form a final consensus, mainly by converting subjective opinions into objective results.

There are many variations to the delphi method, and we have chosen a modified approach that we have collectively agreed would be most reliable. The main data collection is by structured questionnaires through online means, for flexibility. The experts must have deep knowledge on the subject and years of experience in digital forensics, either as a researcher or an investigator. The content of the questionnaire contains the analysis of the data as well as raw statistics for the expert’s own interpretation; each round of MCQ will have a corresponding anonymous discussion between the experts. In total, three rounds of questioning will determine the preferred choice, with all results and opinions extensively documented.

Chapter 7

Expected Outcomes

The study anticipates the following outcomes, tied to your four objectives:

1. **Evidence base for standardisation (Objective 1):** A categorised synthesis of procedural variations, documentation gaps and reported limitations across at least 50 peer-reviewed studies, confirming and extending the gap observed in the 40-report analysis.
2. **A unified forensic protocol (Objective 2):** A phase-by-phase protocol with at least 40 discrete steps and mandatory documentation requirement, built on NIST (National Institute Standards and Technology), ACPO (Association Chief of Police Officers) and ISO/IEC 27037, and validated by at least five certified practitioners (target of at least 80% consensus).
3. **A reliability metric (Objective 3):** A scoring system with empirically derived weightings, tested on at least 100 historical cases, with inter-rater reliability of Cohen's $\kappa \geq 0.80$.
4. **Empirical evidence of effectiveness (Objective 4):** Across at least 20 simulated scenarios, statistically significant ($p < 0.05$) improvement in reproducibility and reduced error rates compared with ad hoc methods, along with measured latency.

Significance: For researchers, the outcomes provide a reproducible, measurable baseline for comparing forensic methods. For practitioners, they provide a repeatable procedure with documentation requirements. For courts, they provide a transparent, quantifiable basis for assessing evidentiary weight, which supports admissibility reasoning under the Daubert criteria.

Deliverables: the protocol document, the reliability scoring framework and its validation results, the experimental dataset and scripts, and a research article summarising the findings.

Chapter 8

Project Timeline

In terms of the timeline for the literature review report and research project proposal, Figure 8.1 below showcases the timeline of events in a gantt chart.

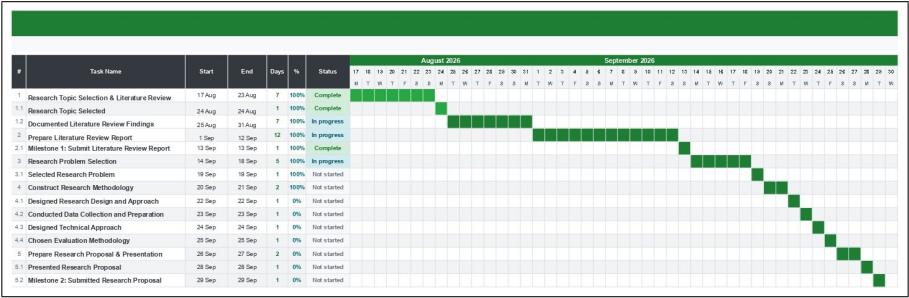


Figure 8.1: Gantt Chart for Literature Review Report and Project Proposal

For the expected timeframe for each section of the Research Project itself, Figure 8.2 below showcases the estimated timeline of events for the Research Project. It is to note that the milestones for the project is highlighted red while the checkpoints for the project is highlighted blue.

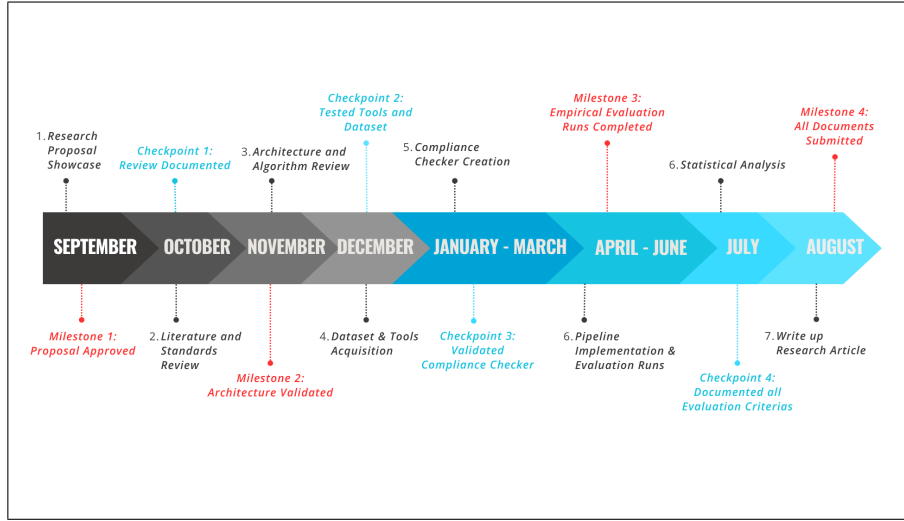


Figure 8.2: Project Timeline for our Research Project

8.1 Conclusion

Summary of contributions: This proposal aims to respond to a documented gap: digital forensics lacks standardised procedures and quantifiable reliability measures, which weakens the reproducibility and legal defensibility of digital evidence. We propose a unified forensic protocol, a reliability metric for scoring digital evidence, and a controlled empirical evaluation that tests both against ad hoc baselines using measurable criteria such as Cohen’s kappa, error rate, F1-score and chain-of-custody integrity. Together these aim to replace subjective, inconsistent practice with a repeatable and testable approach.

Final remarks: As cybercrime grows more complex, courts will inevitably have to rely more heavily upon digital evidence, which must be trustworthy and open to scrutiny. Grounding digital forensic practices through standardised protocols alongside metrics for the reliability of evidence within legal reports produced from said practices offers a path to making digital evidence more consistent, transparent and defensible in a court of law.

References

- Arif, T., Camacho, D., & Park, J. H. (2025). Unveiling cybersecurity mysteries: A comprehensive survey on digital forensics trends, threats, and solutions in network security. *Journal of Network and Computer Applications*, 104296.
- Carranza, A., & DeCusatis, C. (2021). Software validation and daubert standard compliance of an open digital forensics model. *Journal of Machine Intelligence and Data Science (JMIDS)*, 2.
- Ilyas, A., et al. (2024). The role of digital forensics in criminal investigations: An analysis of electronic evidence in the indonesian legal system. *Pakistan Journal of Criminology*, 16(4), 255.
- Osamor, J., Akerele, I., Lawal, A., Osisami, D., Aniekwe, C., Omoleye, D., Adedamola, A., Olakunle, R., Love, A. D., & Christopher, D. (2026). Comprehensive forensic analysis of a spoofed microsoft email: A nist incident response approach. *2026 International Conference on Signal Analysis for Smart Systems (SIGNASS)*, 1–6.
- Prajwalasimha, S., Shelke, N., Saini, D. K. J. B., Pimpalkar, A., Kumar, G. H., & Shivamma, D. (2025). Ai-powered defense against advanced persistent threats (apts): Techniques, case studies, and future research directions. *2025 3rd International Conference on Sustainable Computing and Data Communication Systems (ICSCDS)*, 826–831.
- Sanna, S. L., Regano, L., Maiorca, D., & Giacinto, G. (2025). Improving cybercrime detection and digital forensics investigations with artificial intelligence. *arXiv preprint arXiv:2510.14638*.
- Sharma, S. (2024). Digital forensics: Legal standards and practices in cybercrime investigation. *2024 4th International Conference on Innovative Practices in Technology and Management (ICIPTM)*, 1–6.